Blast from the Past - The Derivative of $\ln(x)$

Let $y = \ln(x)$. Solve this equation for x.

Use implicit differentiation to find $\frac{dy}{dx}$ in terms of y.

Use the above answers to show that $\frac{d}{dx}\ln(x) = \frac{1}{x}$.

Some Stuff from Last Time...

Last time, we looked at inverse trig functions. Today, we will differentiate them...

Question

- What are the domain and range of $\arctan x$?
- If $y = \arctan x$, then $_____ = x$.

Question Starting from the identity $\sin^2 y + \cos^2 y = 1$, show that $\tan^2 y + 1 = \sec^2 y$.

Differentiating...

tan x

Let $y = \tan x$. Use implicit differentiation on the equation from the second bullet point above to find $\frac{dy}{dx}$ in terms of y.

Use the identity you showed above to show that $\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$.

sin x and cos x

Use a similar process to the above, this time using the identity $\sin^2 y + \cos^2 y = 1$ directly to find the derivatives of $\sin^{-1} x$ and $\cos^{-1} x$.

Record your results here:

$\frac{d}{dx}\tan^{-1} x =$ _____ $\frac{d}{dx}\sin^{-1} x =$ _____ $\frac{d}{dx}\cos^{-1} x =$ _____

Questions

1. (a) What is $\sin(\arcsin x)$?

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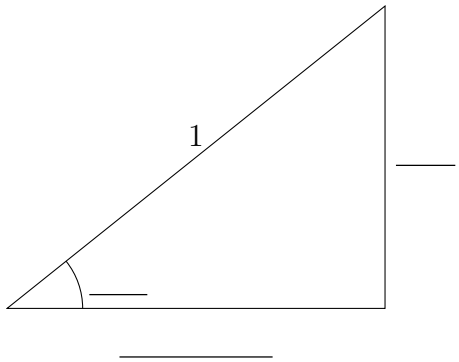
(b) Given the above, what should $\frac{d}{dx}\sin(\arcsin x)$ be?

(c) Use the chain rule instead to differentiate $\sin(\arcsin x)$.

(d) Given your last two answers, what must $\cos(\arcsin x)$ be equal to? Explain.

(e) Let $y = \arcsin x$. What kind of object is y? An angle? A side length? Something else? Label x and y on the triangle below.

(f) Find the length of the missing side and explain why this proves your answer in (d).



2. Find the derivatives of the following functions:

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(a) $p(t) = t\sin^{-1} t$

(b) $f(x) = \frac{\arctan x}{x}$

3. At what value(s) of x does the graph of $\sin^{-1} x$ have vertical tangents?
□Regrade (Note: show this using the derivative of $\sin^{-1} x$, but recall that we also stated the answer last time when we drew the graph of $\sin^{-1} x$. Can you recall how we got there?)

4. Use the derivative of $\tan^{-1} x$ to show that $\tan^{-1} x$ has horizontal asymptotes as $x \to \infty$
□Regrade and as $x \to -\infty$. (Hint: If a function has a horizontal asymptote, what must be true of its derivative as $x \to \infty$ or $-\infty$?)

5. Use a linear approximation to estimate arctan(1.1). (Hint: What is arctan(1)?) Is your
□Regrade approximation an overestimate of the true value of arctan(1.1) or an underestimate of it?